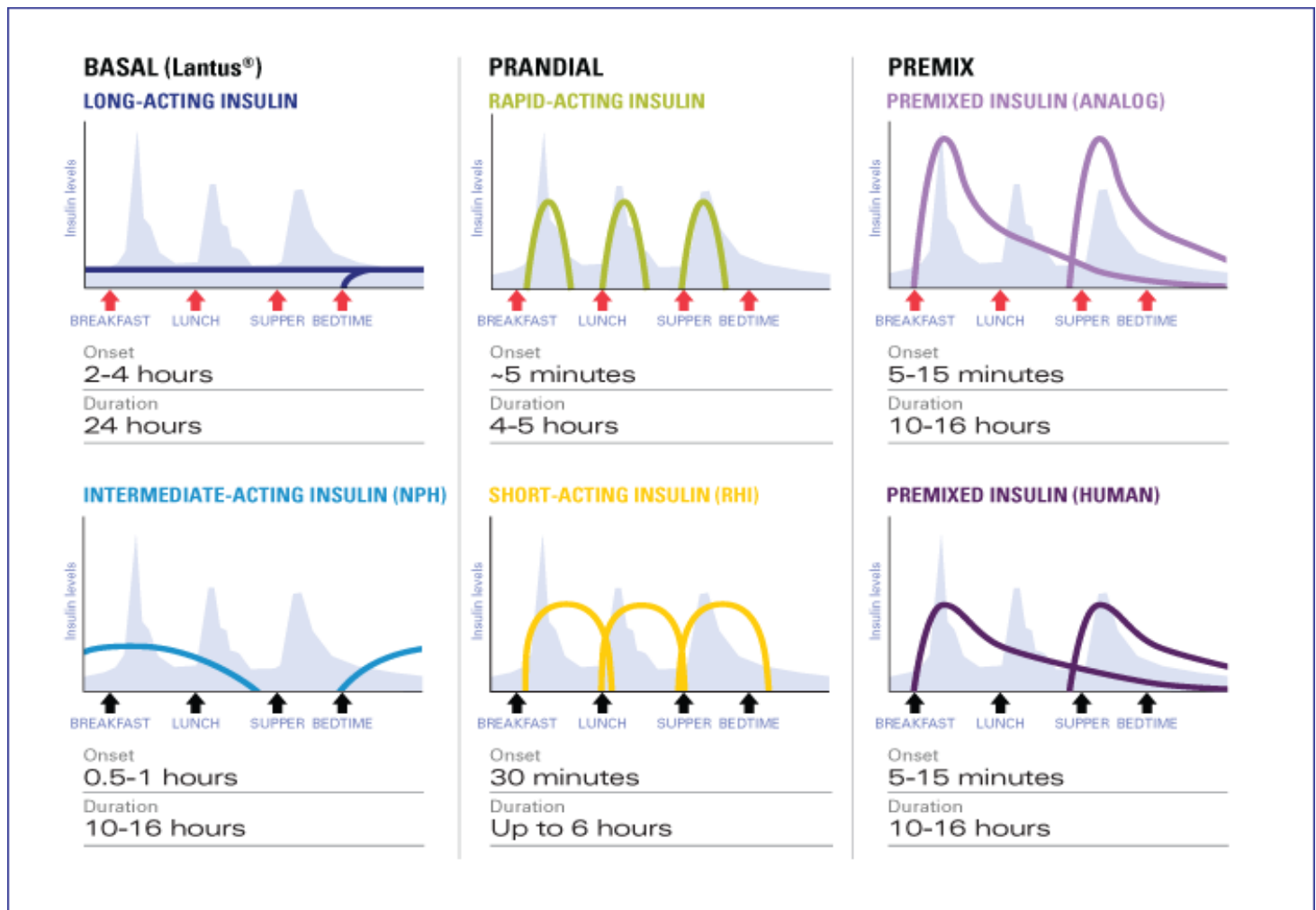


The principles of insulin adjustment guidance

Tips for insulin titration

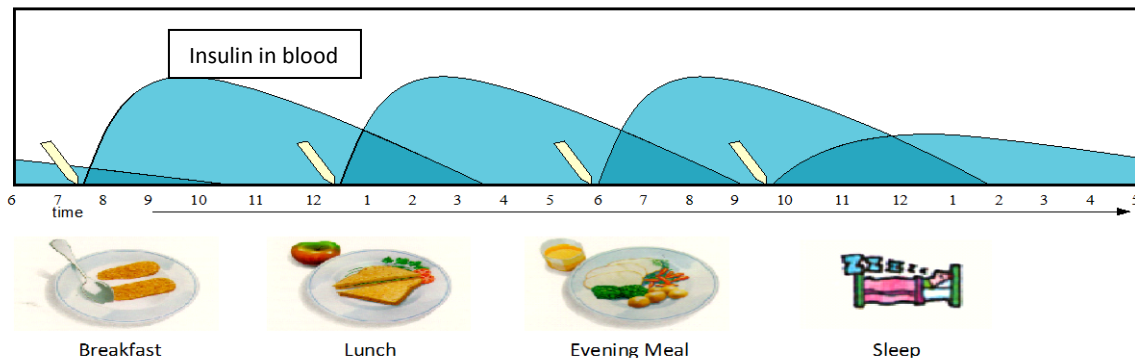
- Blood glucose (BG) monitoring is needed to help identify the efficacy of treatment in diabetes.
- Monitor blood glucose according to individual need. Measurement of blood glucose before meals and two hours following a meal can provide important information to adjust insulin. The recommended target BG level in in-patients with diabetes is 6-10 mmol/L with 4-12 mmol/L being acceptable (Joint British Diabetes Society). A guideline for the management of post meal glucose http://www.idf.org/webdata/docs/Guideline_PMG_final.pdf
- Insulin onset and time action profiles are demonstrated below (figure 1):
- Rapid acting analogues and soluble insulin should be prescribed in relation to meal times
- Rapid acting analogue insulin preparations can be prescribed and administered before, with or after a meal.
- Premixed insulin preparations e.g. Novomix 30 and Humulin M3 are normally prescribed before breakfast and before an evening meal.
- Intermediate acting insulin e.g. Insulatard and Humulin I can be given before breakfast and evening meal or bedtime (e.g. 22 00hour) as basal insulin.
- Basal insulin preparations e.g. Lantus and Levemir can be administered outwith mealtimes. Basal insulin preparations can be prescribed once or twice daily (breakfast and bedtime).
- STAT doses of rapid acting or soluble insulin to 'correct' hyperglycaemia should be used with caution as this can precipitate hypoglycaemia.
- Patients prescribed insulin can be referred for education e.g. to facilitate insulin dose adjustment in relation to the carbohydrate content of food, exercise regimens, illness.
- Personal insulin sensitivity can be calculated to identify how much short acting insulin is required to reduce BG e.g. 1 unit: 2mmol/L.

Figure 1 Schematic diagram of insulin onset and time action profiles



Basic Guidelines - Insulin Adjustment for Multiple Daily Injection Regimens

Short acting insulin preparations e.g. Actrapid, Humulin S, Novorapid, Humalog, Apidra
Intermediate acting insulin e.g. Humulin I, Insulatard, Insuman Basal



Insulin adjustment for hyperglycaemia (Blood Glucose (BG) > 10mmol/L)

THINK

- The recommended target BG level in in-patients with diabetes is 6-10 mmol/L with 4-12 mmol/L being acceptable (Joint British Diabetes Society)
- Some patients for example pregnant women will have lower BG targets.
- Adjusting insulin reactively to 'quick fix' high blood glucose readings may precipitate hypoglycaemia.

CHECK

- Ideally, review the pattern of blood glucose levels over the previous 48 hours.
- If the trend of the blood glucose is elevated above 8 mmol/L:

ACT

- Adjust insulin dose prior to the elevated blood glucose level as follows:
 - If BG is elevated at lunchtime, increase the breakfast insulin dose.
 - If BG is elevated at teatime, increase the lunchtime insulin dose.
 - If BG is elevated at suppertime, increase the evening meal insulin dose.
 - If BG is elevated at breakfast, time increase basal insulin dose.
 - Basal insulin preparations include Insulatard, Humulin I Lantus, Levemir).
- An increase of 10 % of the dose of insulin is generally recommended.
- Check ketones in people with type 1 diabetes and all patients during acute illness.
- Observe the pattern of premeal blood glucose levels thereafter and titrate insulin again if necessary.

Insulin adjustment for hypoglycaemia (BG <4mmol/L)

THINK

- Hypoglycaemia can cause unpleasant symptoms and patient harm.
- Hypoglycaemia can lead to increased length of hospital stay.

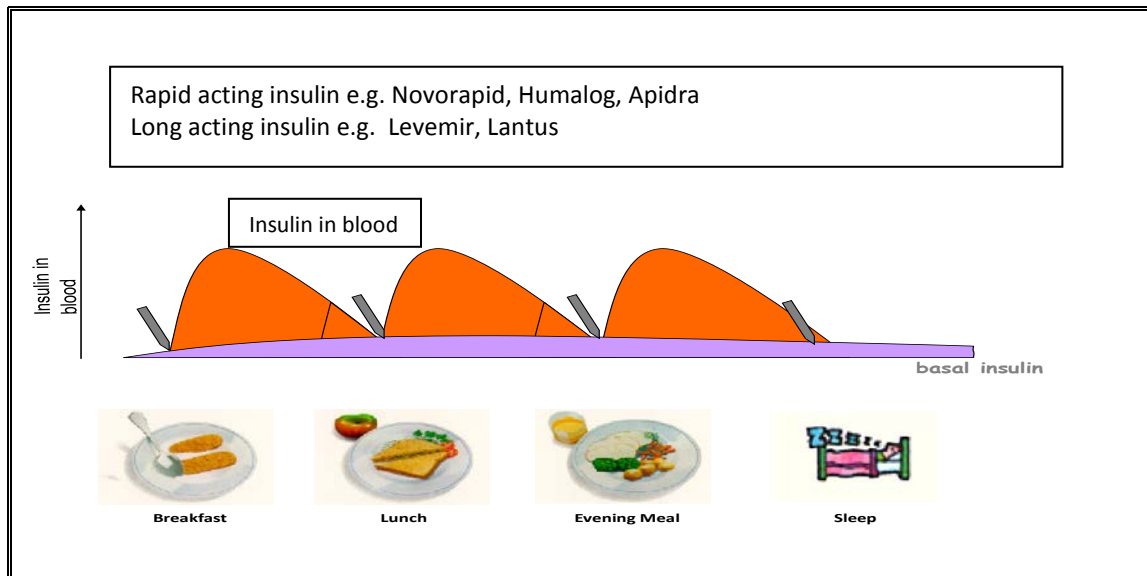
CHECK

- Review pattern of blood glucose levels over the previous 48 hours.

ACT

- Treat hypoglycaemia promptly and recheck BG to confirm recovery/identify need for further treatment
- In the event of an episode of **unexplained hypoglycaemia** (BG <4mmol/L) reduce insulin.
- If the BG is less than 6 mmol/L consider reduction in insulin, if BG < 4mmol/L reduce insulin:
 - If BG < 4 mmol/L before lunch, consider **reduction** in the breakfast insulin dose
 - If BG is < 4 mmol/L before the evening meal, consider **reduction** in the lunchtime insulin dose.
 - If BG is < 4 mmol/L before suppertime, consider **reduction** the evening meal insulin dose.
 - If BG is < 4 mmol/L before breakfast time, consider reduction in the basal insulin dose.
 - Basal insulin preparations include Insulatard, Humulin I Lantus, Levemir.
- A reduction of 10 % of the dose of insulin is generally recommended.
- Observe the pattern of the blood glucose levels thereafter and titrate insulin again if necessary.
- **DO NOT OMIT INSULIN**; ask for advice from diabetes team if necessary.
- If blood glucose level is low (below 4mmol/L) and insulin injection is due, provide patient with 15-20 grams of quick acting carbohydrate to increase blood glucose level, then administer insulin and meal as usual.
- Address the cause of hypoglycaemia and review treatment aiming to prevent it from happening again.

Basic Guidelines - Insulin Adjustment for Multiple Daily Injection Regimens



Insulin adjustment for hyperglycaemia (blood glucose > 10mmol/L)

THINK

- The recommended target BG level in in-patients with diabetes is 6 -10 mmol/L with 4-12 mmol/L being acceptable (Joint British Diabetes Society).
- Some patients for example pregnant women will have lower BG targets.
- Adjusting insulin reactively to 'quick fix' high blood glucose readings may precipitate hypoglycaemia.

CHECK

- Ideally, review the pattern of blood glucose levels over the previous 48 hours.
- If the trend of the blood glucose is elevated above 8 mmol/L:

ACT

- Adjust insulin dose prior to the elevated blood glucose level as follows:
 - If BG is elevated at lunchtime, increase the breakfast insulin dose.
 - If BG is elevated at teatime, increase the lunchtime insulin dose.
 - If BG is elevated at suppertime, increase the evening meal insulin dose.
 - If BG is elevated at breakfast time, increase basal insulin dose.
 - Basal insulin preparations include Insulatard, Humulin I Lantus, Levemir).
- An increase of 10 % of the dose of insulin is generally recommended.
- Check ketones in people with type 1 diabetes and all patients during acute illness.
- Observe the pattern of premeal blood glucose levels thereafter and titrate insulin again if necessary

THINK

- **Insulin adjustment for hypoglycaemia (blood glucose <4mmol/L)**
- Hypoglycaemia can cause unpleasant symptoms and patient harm
- Hypoglycaemia can lead to increased length of hospital stay

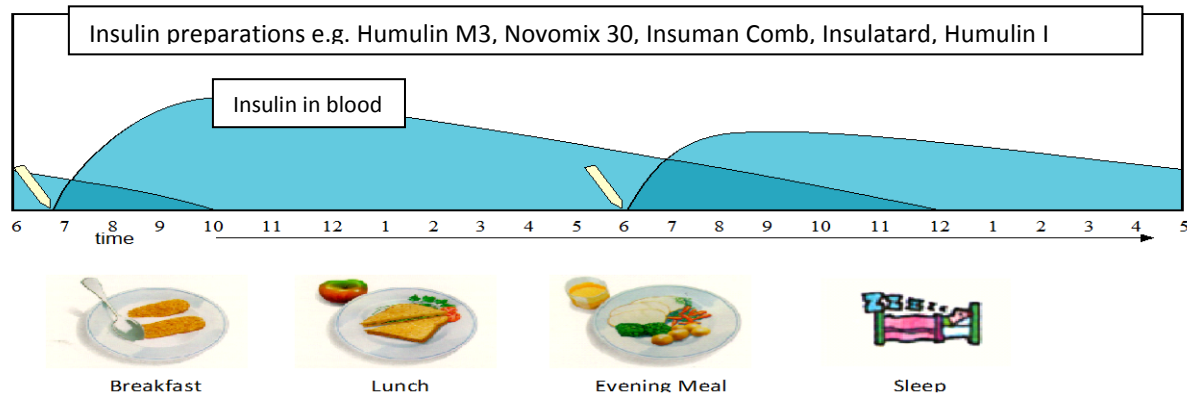
CHECK

- Review pattern of blood glucose levels over the previous 48 hours

ACT

- Treat hypoglycaemia promptly and recheck BG to confirm recovery/identify need for further treatment.
- In the event of an episode of **unexplained hypoglycaemia** (BG <4mmol/L) reduce insulin.
- If the BG is less than 6 mmol/L consider reduction in insulin, if BG < 4mmol/L reduce insulin:
 - If BG is < 4 mmol/L before lunch, reduce the breakfast insulin dose.
 - If BG is < 4 mmol/L before the evening meal, reduce the lunchtime insulin dose.
 - If BG < 4mmol/L before suppertime, reduce the evening meal insulin dose.
 - If BG < 4 mmol/L before breakfast time, reduce the basal insulin dose.
 - Basal insulin preparations include Insulatard, Humulin I Lantus, Levemir.
- A reduction of 10 % of the dose of insulin is generally recommended.
- Observe the pattern of the blood glucose levels thereafter and titrate insulin again if necessary.
- **DO NOT OMIT INSULIN;** ask for advice from diabetes team if necessary.
- If blood glucose level is low (below 4mmol/L) and insulin injection is due, provide patient with 15-20 grams of quick acting carbohydrate to increase blood glucose level, and then administer insulin and meal as usual.
- Address the cause of hypoglycaemia and review treatment aiming to prevent it from happening again.

Basic Guidelines - Insulin Adjustment for Twice Daily Insulin Regimen



Insulin adjustment for hyperglycaemia (blood glucose > 10mmol/L)

THINK

- The recommended target BG level in in-patients with diabetes is 6 -10 mmol/L with 4-12 mmol/L being acceptable (Joint British Diabetes Society).
- Some patients for example pregnant women will have lower BG targets.
- Adjusting insulin reactively to 'quick fix' high blood glucose readings may precipitate hypoglycaemia.

CHECK

- Ideally, review pattern of blood glucose levels over the previous 48 hours.
- If the trend of the blood glucose is elevated:

ACT

- Adjust insulin dose prior to the elevated blood glucose level as follows:
- If the blood glucose is elevated lunch and tea, increase the breakfast insulin dose.
- If the blood glucose is elevated before bed and before breakfast, increase the teatime insulin dose.
- An increase of 10% of the dose of insulin is generally recommended.
- Check ketones in people with type 1 diabetes and all patients during acute illness.
- Observe the pattern of the premeal blood glucose levels thereafter and adjust the insulin dose again if necessary.
- Occasionally a different 'mixture' of insulin is required e.g. change from Humulin I to Humulin M3.

THINK

- **Insulin adjustment for hypoglycaemia (blood glucose <4mmol/L)**
- Hypoglycaemia can cause unpleasant symptoms and patient harm.
- Hypoglycaemia can lead to increased length of hospital stay.

CHECK

- Review pattern of blood glucose levels over the previous 48 hours.

ACT

- Treat hypoglycaemia promptly and recheck BG to confirm recovery/identify need for further treatment.
- In the event of an episode of **unexplained hypoglycaemia** (BG <4mmol/L) reduce insulin.
- **If the BG is less than 6 mmol/L consider reduction in insulin, if BG < 4mmol/L reduce insulin:**
- If BG is < 4 mmol/L before lunch and/or before tea, reduce breakfast insulin.
- If BG is < 4 mmol/L before bed and/or before breakfast, reduce the evening meal insulin dose.
- A reduction of 10 % of the dose of insulin is generally recommended.
- Observe the pattern of the blood glucose levels thereafter and titrate insulin again if necessary.
- **DO NOT OMIT INSULIN;** ask for advice from diabetes team if necessary.
- If BG level is low (below 4mmol/L) and insulin injection is due, provide patient with 15-20 grams of quick acting carbohydrate to increase blood glucose level, and then administer insulin and meal as usual
- Address the cause of hypoglycaemia and review treatment aiming to prevent it from happening again.

Diabetes information www.diabetes-healthnet.ac.uk

Guideline for Monitoring and Managing Glycaemic Control for Inpatients with Diabetes Blood glucose target in patients with diabetes in hospital is 6 -10 mmol/L with 4-12mmol/L being acceptable. Assess target blood glucose target range in frail or elderly patients to reduce the risk of hypoglycaemia. In women during pregnancy blood glucose levels should be agreed to reduce the risk of problems with fetal growth and neonatal hypoglycaemia. Frequency of blood monitoring should be assessed for each individual. Check ketones at diagnosis of diabetes and consider checking if BG is greater than 15mmol/L. Check ketones in patients who are acutely unwell. Check ketones in pregnant women who are acutely unwell <i>irrespective</i> of BG level. The blood ketone meter range is 0 – 8 mmol/L. A ketone level above 0.6 mmol/L is abnormal.	
Guideline for Hyperglycaemia Management	Guideline for Hypoglycaemia Management
THINK Blood glucose (BG) levels > 10mmol/L increase risk of osmotic symptoms of diabetes, dehydration and can delay healing. Hyperglycaemia can lead to Diabetic Ketoacidosis (DKA) or Hyperosmolar Hyperglycaemia Syndrome (HHS). Consider the causes of high blood glucose e.g. • Infection and/or stress response to illness • Steroid therapy • Nutrition e.g. supplements, NG feeding or dietary indiscretion • Insulin and /or diabetes medication omission/inadequate dose • Insulin or drug administration at an inappropriate time • Insulin absorption problem e.g. technique /administration/injection site Pancreatic insufficiency / pancreatitis	THINK Hypoglycaemia i.e. blood glucose (BG) level < 4mmol/L is a potentially dangerous side effect of insulin therapy and hypoglycaemic agents e.g. Gliclazide, Glipizide, Glimepiride, Glibenclamide. Hypoglycaemia must be avoided and if present, prompt treatment is required – see recommendation below. Consider the causes of low blood glucose levels e.g. • Inadequate carbohydrate food intake • Too much insulin and /o oral hypoglycaemic medication • Reduction or withdrawal of steroid therapy • Insulin absorption problems e.g. technique / administration /injection site • Increased activity • Renal or hepatic impairment • Pancreatic insufficiency
CHECK Identify potential cause(s) of elevated BG levels i.e. • Assess pattern of BG levels e.g. over previous 48hrs • Check for signs of infection • Check insulin/medication prescription, dose, time of administration, food intake, activity • Check for factors which may affect insulin absorption • Check credibility of BG monitoring e.g. hand washing prior to testing • Check ability to self manage medication • Check insulin delivery device • Check for ketones during acute illness/vomiting and in all pregnant patients • Ensure that patients using Continuous Subcutaneous Insulin Infusion (CSII) check pump function, pump programming, infusion set and its site	CHECK Following identification and treatment of hypoglycaemia: • Assess pattern of BG levels e.g. over previous 48 hours • Assess recent nutritional status • Identify the drugs prescribed that may precipitate hypoglycaemia • Have steroids been withdrawn or reduced recently? • Check insulin/medication prescription, dose, time of administration, food intake, activity • Check for factors which may affect insulin absorption • Check ability to self manage medication if appropriate • Establish cause of hypoglycaemia and review medication • Increase frequency of BG monitoring following treatment change
ACT Address the cause(s) of hyperglycaemia • If the trend of the pre-meal BG level is > 10mmol/l, review medication, and clinical status and adjust treatment. • If ketone level is > 0.6 mmol/L refer for urgent medical review • If ketone level is > 0.6 mmol/L increase fluid intake and treat with insulin • Review and check BG and ketones 2-4 hourly until confirmed ketone free • Consider adjustment of insulin /medication if steroid therapy is prescribed • Increase frequency of BG monitoring following treatment change • Further adjust insulin/medication on an ongoing basis if necessary • Inform and agree medication change with patient/parent/carer • Refer to the Diabetes Team for advice as required	ACT Treat hypoglycaemia immediately with 15 - 20 gram of quick acting carbohydrate • If patient is able to swallow - administer 60 mL of Glucojuice or 90 - 120mL Lucozade® • If patient is confused or drowsy but able to swallow: administer 1-2 tubes of Glucogel • If patient is unconscious/unable to swallow: administer IV Glucose 10% 150 mls or 20% 75 ml or 1mg IM Glucagon (adults) • Note Glucagon is not suitable in malnourished patients, in severe liver disease, or those on oral hypoglycaemic agents • Provide complex carbohydrate snack promptly e.g. wholemeal bread/toast • Observe and chaperone patient until recovery is complete • Recheck BG in 15 minutes and repeat treatment if necessary • Do not omit insulin: treat the 'hypo' and administer the insulin as prescribed • Take appropriate action to prevent further hypoglycaemia • Inform and agree medication change with patient/parent/carer • Provide appropriate patient education • Refer to the Diabetes Team for advice as required